WORKSHEET 8.10

Unit 8 Synthesis: The Area and Volume Free-Response Question

Unit 8: Applications of Integration

Key ideas

- The usual parts: an area, a volume of revolution, and a volume with known cross-sections, all for one region.
- Find the points of intersection first, and store them in the calculator. Do not round until the final answer.
- Write every integral in full, with limits, before giving its value. The set-up earns most of the points.
- "Write, but do not evaluate" means stop at the integral.

Worked examples

Example 1 · The area

Let R be the region enclosed by $y = \sqrt{x}$ and $y = \dfrac{x}{3}$. Find the area of R.

Solution. They meet at $x = 0$ and $x = 9$: $\displaystyle\int_0^9 \left(\sqrt{x} - \tfrac{x}{3}\right)\,dx = 18 - \frac{27}{2}$.

$\dfrac{9}{2}$.

Example 2 · A solid of revolution

Find the volume when R is revolved about the x-axis.

Solution. $\pi \displaystyle\int_0^9 \left(x - \tfrac{x^2}{9}\right)\,dx = \pi\left(\frac{81}{2} - 27\right)$.

$\dfrac{27\pi}{2}$.

Example 3 · Known cross-sections

R is the base of a solid whose cross-sections perpendicular to the x-axis are squares. Find the volume.

Solution. $\displaystyle\int_0^9 \left(\sqrt{x} - \tfrac{x}{3}\right)^2\,dx = \frac{81}{2} - \frac{324}{5} + 27$.

$\dfrac{27}{10}$.

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Practice

A calculator may be used where a decimal answer is requested. Give decimals to three places.

CORE

1. Let S be the region enclosed by $y = x^2$ and $y = 3x$. Find the area of S.

2. Write, but do not evaluate, an integral for the volume when S is revolved about the x-axis.

3. Evaluate the integral in Question 2.

4. S is the base of a solid whose cross-sections perpendicular to the x-axis are squares. Write, but do not evaluate, an integral for the volume.

EXAM LEVEL

5. Let T be the region enclosed by $y = 4 - x^2$ and $y = e^x$. Find the x-coordinates of the two points of intersection.

6. Find the area of T.

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7. Find the volume of the solid generated when T is revolved about the x-axis.

8. T is the base of a solid whose cross-sections perpendicular to the x-axis are semicircles with diameter in the base. Find the volume.

Question 9 · Free-response practice

Let R be the region enclosed by $y = 2\sqrt{x}$ and $y = x$.

(a) Find the area of R.

(b) Find the volume of the solid generated when R is revolved about the x-axis.

(c) R is the base of a solid whose cross-sections perpendicular to the y-axis are semicircles with diameter in the base. Find the volume.

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Worked solutions

1. $\displaystyle\int_0^3 (3x - x^2)\,dx = \frac{9}{2}$.

2. $\displaystyle\pi\int_0^3 (9x^2 - x^4)\,dx$.

3. $\displaystyle\pi\left(81 - \frac{243}{5}\right) = \frac{162\pi}{5}$.

4. $\displaystyle\int_0^3 (3x - x^2)^2\,dx$.

5. $x \approx -1.965$ and $x \approx 1.058$.

6. $\displaystyle\int_{-1.965}^{1.058} (4 - x^2 - e^x)\,dx \approx 6.428$.

7. $\displaystyle\pi\int_{-1.965}^{1.058} \left[(4 - x^2)^2 - e^{2x}\right]dx \approx 84.705$.

8. $\displaystyle\frac{\pi}{8}\int_{-1.965}^{1.058} \left(4 - x^2 - e^x\right)^2 dx \approx 6.422$.

9. (a) They meet at $x = 0$ and $x = 4$: $\displaystyle\int_0^4 (2\sqrt{x} - x)\,dx = \frac{32}{3} - 8 = \frac{8}{3}$.

   (b) $\displaystyle\pi\int_0^4 (4x - x^2)\,dx = \pi\left(32 - \frac{64}{3}\right) = \frac{32\pi}{3}$.

   (c) At height y the region runs from $x = \dfrac{y^2}{4}$ to $x = y$: $\displaystyle\frac{\pi}{8}\int_0^4 \left(y - \tfrac{y^2}{4}\right)^2 dy = \frac{\pi}{8}\cdot\frac{32}{15} = \frac{4\pi}{15}$.

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